ARIS

TECHNOLOGY ADOPTION INTELLIGENCE

ADOPTION DECISION BRIEF
2026-05-31

Adoption Decision Brief

langchain · building a RAG pipeline in Python

TRIAL

64.8 / 100

CONFIDENCE 96%

LangChain is a fast-moving, widely adopted framework for LLM applications that carries real operational cost. Velocity is high and the Python RAG ecosystem around it is deep — yet a low **bus factor**, one **unpatched critical** CVE, and thin evidence of large-scale production reliability keep this short of an outright ADOPT. Pilot it behind an abstraction layer first.

+ ecosystem maturity · + stack compatibility · − production adoption · − security

DIMENSION SCORES

Dimension	Score
Maintenance Health commits · bus factor · issues · releases	65
Security Risk OSV CVEs · unpatched only	55
Stack Compatibility Python RAG ecosystem fit	75.5
Ecosystem Maturity download momentum · alternatives	75.5
Production Adoption confirmed enterprise case studies	50
Learning Curve docs · setup · debugging friction	68.5

RECOMMENDATION RATIONALE

TRIAL reflects a strong-but-uneven profile: ecosystem maturity and stack fit score well (75.5), while production adoption (50) and security (55) pull the weighted total to 64.8. It moves to **ADOPT** if the unpatched critical is resolved and two named enterprises with scale data appear; it falls to **HOLD** if release cadence slips past 180 days.

ALTERNATIVES TO CONSIDER

For Python RAG, **LlamaIndex** is the most-cited alternative when retrieval is the core workload; **Haystack** suits teams wanting an opinionated, production-first pipeline; direct orchestration over provider SDKs is common once teams outgrow the abstraction. LangChain wins on breadth and prototyping speed; it loses when a small, debuggable surface matters more.

CAVEATS & DATA GAPS

Confidence is 96% because the deterministic signals — GitHub health and OSV CVE data — were fully available. Production adoption is the weakest evidence (one enterprise, no scale data); stack compatibility used the ecosystem-maturity fallback.

MAINTAINER BURNOUT	ABANDONMENT PROB.	TOTAL CVES	UNPATCHED CRITICAL
MEDIUM	0.20	37	1

HOW THIS SCORE WAS COMPUTED

Six weighted dimensions, each scored 0–100 by deterministic code — no model produces a number. The weighted total maps to the verdict band.

DIMENSION	WEIGHT	PRIMARY SOURCE	SCORE
Maintenance health	20%	GitHub commits, bus factor, issue close-rate, release cadence	65
Security risk	20%	OSV CVE feed — unpatched severities only	55
Stack compatibility	20%	Target-repo dependency match (ecosystem fallback)	75.5
Ecosystem maturity	15%	Download momentum + production + alternatives count	75.5
Production adoption	15%	Confirmed enterprise case studies from web search	50
Learning curve	10%	Community-reported docs, setup, debugging friction	68.5

VERDICT BANDS

ADOPT ≥ 75
TRIAL ≥ 60
HOLD ≥ 40
AVOID < 40

SECURITY VETO

A security score below 30 caps the verdict at **HOLD** regardless of the weighted total. Live attack surface overrides a strong average.

CONFIDENCE

Data completeness + deterministic coverage − contradiction count. Missing data lowers *confidence*, never the score.